into compact group at the equator of the cell is known as **metaphase plate**. This is the stage where the chromosomal morphology can be easily studied.

Kinetochore is a DNA–Protein complex present at the centromere where microtubules are attached. It is a trilaminar disc like plate.

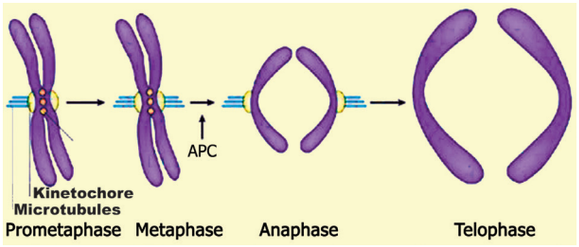


Figure 7.5: Anaphase promoting complex cyclosome

Anaphase

Each chromosome splits simultaneously and two daughter chromatids begin to migrate towards two opposite poles of a cell. Each centromere splits longitudinally into two, freeing the two sister chromatids from each other. When sister chromatids separate the actual partitioning of the replicated genome is complete.

APC (Anaphase Promoting Complex) is a cluster of proteins that induces the breaking down of cohesion proteins which leads to the separation of chromatids during mitosis (Figure 7.5). Thus it helps in the transition of metaphase to anaphase.

Telophase

Two sets of daughter chromosomes reach opposite poles of the cell and mitotic spindle disappears. Division of genetic material is completed during karyokinesis, followed by cytokinesis (division of cytoplasm). Nucleolus and nuclear membranes reforms. Nuclear membrane form around each set of **chromosomes**. Now the chromosomes decondense. In plants, phragmoplast are formed between the daughter cells. Cell plate is formed between the two daughter cells, reconstruction of cell wall takes place. Finally cells are separated by the distribution of organelles, macromolecules into two newly formed daughter cells.

7.3.4 Cytokinesis

Cytokinesis in Animal Cells

It is a contractile process. The ring consists of a bundle of microfilaments assembled from **actin** and **myosin**. This fibril generates a contractile force, that draws the ring inward forming a cleavage furrow in the cell. Thus it divides the cell into two.

Check your grasp!
What effect does mitosis have on transcription?
During mitosis transcription stops.

Cytokinesis in Plant Cell

Division of the cytoplasm often starts during telophase. In plants, cell plate grows from centre towards lateral walls.

Phragmoplast contains microtubules, actin filaments and vesicles from golgi apparatus and ER. Microtubule of the pharagmoplast move to the equator, fuse to form a new plasma membrane and the materials which are placed

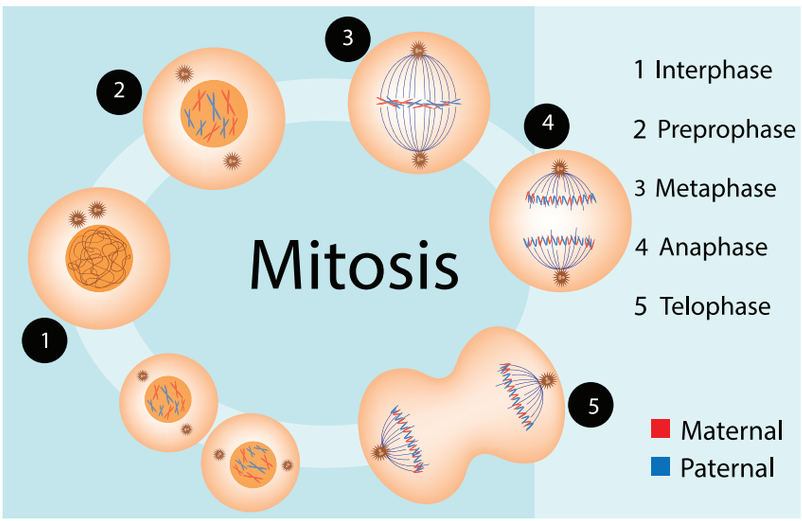


Figure 7.6: Mitosis

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